Gradient Boosting Regression for NBA Player Prop Prediction:

Walk-Forward Validation, Feature Attribution, and Simulated Betting Performance

Abba Otieno Ndomo

[Institution]

abba.ndomo@gmail.com

May 11, 2026

Abstract

Player proposition (prop) betting markets in the NBA present a challenging regression problem: predicting individual player statistics — points, rebounds, and assists — from historical performance and contextual features. We develop a gradient boosting regression system that trains separate XGBoost models for each statistic using per-player rolling averages, exponential moving averages, season baselines, hot/cold streak indicators, and opponent defensive context features. Trained on 1.1 million player-game observations spanning 2016–2024, the models are evaluated under walk-forward cross-validation across four held-out NBA seasons (2021–2024), yielding mean absolute errors of 4.84 points, 1.97 rebounds, and 1.41 assists. We compare performance against three baselines — a naive recency heuristic, linear regression, and random forest — all trained on identical feature sets. SHAP (SHapley Additive exPlanations) analysis identifies the 10-game rolling average, season expanding mean, and opponent defensive rolling average as the most influential predictors. A simulated betting experiment using synthetic market lines demonstrates that the XGBoost model generates positive return on investment relative to a linear regression baseline (+13.1% ROI on points at a 1-point edge threshold, N = 17,685 bets), suggesting the model captures non-linear structure beyond what simpler models extract from the same features. We discuss limitations of synthetic line evaluation and directions for future work using historical closing lines.

Keywords: NBA, player proposition betting, gradient boosting, XGBoost, SHAP, walk-forward cross-validation, sports analytics

Contents

1 Introduction 5

2 Related Work 6

2.1 Basketball Outcome Prediction 6

2.2 Sports Betting Market Efficiency 6

2.3 Gradient Boosting for Sports Prediction 6

2.4 Explainability in Sports Models 6

3 Data and Feature Engineering 6

3.1 Dataset 6

3.2 Per-Player Rolling Features 7

3.3 Opponent Defensive Context 7

3.4 Feature Sets by Statistic 8

4 Methodology 9

4.1 Model Architecture 9

4.2 Walk-Forward Cross-Validation 9

4.3 Baseline Models 9

4.4 SHAP Feature Attribution 10

4.5 Betting Simulation 10

5 Results 11

5.1 Prediction Accuracy 11

5.2 SHAP Feature Importance 12

5.3 Betting Simulation 12

6 Limitations and Future Work 13

7 Conclusion 14

A Per-Fold MAE Results **15**

B SHAP Feature Importance Tables **15**

C Full Betting Simulation Results **15**

1 Introduction

The NBA player proposition (prop) betting market has grown substantially following the 2018 repeal of the Professional and Amateur Sports Protection Act (PASPA), with per-game player prop volume now rivalling traditional moneyline and spread markets [**?**]. Sportsbooks set prop lines by estimating expected statistical output — typically points, rebounds, or assists — for individual players, and bettors wager on whether a player will exceed (*over*) or fall short (*under*) of that line.

The academic literature on basketball prediction has focused predominantly on binary game-outcome classification (win/loss) [**??**]. Individual player stat prediction, which requires regression over a continuous target rather than a binary one, has received comparatively less attention [**?**].

This paper makes the following contributions:

1. A gradient boosting regression pipeline that trains separate XGBoost [**?**] models for points, rebounds, and assists using stat-specific feature sets.

2. A walk-forward cross-validation protocol that evaluates temporal generalization across four held-out NBA seasons (2021–2024), avoiding the look-ahead bias of a single chronological split.

3. Comparison against three baselines — a naive last-5-game average, linear regression, and random forest — all on identical features, isolating the contribution of model complexity.

4. SHAP [**?**] feature attribution identifying which features drive predictions and validating that opponent defensive context adds signal beyond recency.

5. A simulated betting experiment demonstrating statistically robust positive return on investment relative to a linear regression synthetic line, with explicit discussion of

limitations.

2 Related Work

2.1 Basketball Outcome Prediction

? proposed a two-stage XGBoost model for NBA game outcome prediction, identifying lagged shooting percentages, defensive rebounds, and assists as the most predictive features. **?** extended this with SHAP analysis on 2021–2023 NBA data, finding field goal percentage and defensive rebounds to be consistently dominant.

2.2 Sports Betting Market Efficiency

? demonstrated statistically significant inefficiencies in NBA totals lines early in the season, achieving a 56.72% win rate over 20 years by exploiting early-season bias in the closing total.

2.3 Gradient Boosting for Sports Prediction

XGBoost [**?**] has become a standard benchmark on tabular prediction tasks and has been applied successfully to game-level basketball prediction [**??**].

2.4 Explainability in Sports Models

3 Data and Feature Engineering

3.1 Dataset

The primary dataset comprises 1,648,656 player-game records spanning NBA seasons from 1951 to 2025, sourced from [ESPN unofficial API / NBA API]. After filtering to Regular Season and Playoff games with a minimum of 10 minutes played, 1,120,817 records remain across 4,273 unique players. Only seasons from 2016 onward are used for training, reflecting the modern three-point era.

3.2 Per-Player Rolling Features

For each player, the following features are computed using a causal rolling window (shifted by one game to prevent data leakage):

- **Rolling means:** $\bar{x}^{(5)}$, $\bar{x}^{(10)}$ — 5- and 10-game simple moving averages for points, rebounds, assists, minutes, and field goal percentage.
- **Exponential moving average:** $\text{EMA}_t^{(5)} = \alpha x_{t-1} + (1-\alpha)\text{EMA}_{t-1}^{(5)}$, with span $s = 5$ ($\alpha = 2/(s+1)$), capturing recent momentum with exponentially decaying weights.
- **Rolling standard deviation:** $\sigma^{(10)}$ over the last 10 games, used as a dispersion estimate for probability calculation.
- **Days rest:** days since the player's previous game, clipped at 10.
- **Season expanding average:** $\mu_t^{\text{season}} = \frac{1}{t-1}\sum_{k=1}^{t-1} x_k$, computed per player per season year.
- **Hot/cold streak:**

$$\text{hot\_cold} = \frac{\bar{x}^{(5)} - \mu^{\text{season}}}{|\mu^{\text{season}}| + 0.1} \tag{1}$$

  A positive value indicates the player is performing above their season average over the last 5 games.

3.3 Opponent Defensive Context

For each game, player statistics are aggregated at the team level by grouping on (game identifier, opposing team). This yields the total points, rebounds, and assists allowed by each defending team per game. A 10-game rolling average of these totals, shifted by one game (opp_pts_allowed_L10, etc.), is then merged back to each player row as a defensive context feature.

3.4 Feature Sets by Statistic

Each statistic uses a distinct feature set reflecting the different contextual factors that drive scoring, rebounding, and playmaking. Table 1 lists the full feature sets.

Table 1: Feature sets by prediction target.

Stat	**Features**	**Notes**
	is_home, days_rest	Situational
Points	pts_L5/10, pts_ema_L5, pts_std_L10	Recency
	reb_L5, ast_L5, min_L5/10, fg_pct_L5/10	Role context
	season_avg_pts, hot_cold_pts	Baseline / form
	opp_pts_allowed_L10	Opponent defense
	is_home, days_rest	Situational
Rebounds	reb_L5/10, reb_ema_L5, reb_std_L10	Recency
	pts_L5, min_L5/10	Role context
	season_avg_reb, hot_cold_reb	Baseline / form
	opp_reb_allowed_L10	Opponent defense
	is_home, days_rest	Situational
Assists	ast_L5/10, ast_ema_L5, ast_std_L10	Recency
	pts_L5, min_L5/10, fg_pct_L5	Role context
	season_avg_ast, hot_cold_ast	Baseline / form
	opp_ast_allowed_L10	Opponent defense

4 Methodology

4.1 Model Architecture

Three independent XGBoost regressors [?] are trained, one per statistic, using the squared error objective (`reg:squarederror`). Hyperparameters are held constant across all three models: learning rate $\eta = 0.05$, maximum tree depth $= 6$, subsample ratio $= 0.8$, column subsample $= 0.8$, minimum child weight $= 5$. Early stopping with a patience of 50 rounds prevents overfitting, with the test fold used as the validation set during training.

4.2 Walk-Forward Cross-Validation

To evaluate temporal generalization, we employ a walk-forward (rolling-origin) cross-validation scheme [?]. Training data grows by one season per fold, and the model is tested on the immediately following season only:

Table 2: Walk-forward fold definitions.

Fold	**Train**	**Test**
1	2016–2020	2021
2	2016–2021	2022
3	2016–2022	2023
4	2016–2023	2024

4.3 Baseline Models

Four models are compared under identical walk-forward folds:

1. **Naive L5**: predicts the player's last-5-game rolling average. Requires no training and represents a pure recency heuristic.

2. **Linear Regression**: ordinary least squares on the full feature set for each statistic.

3. **Random Forest**: an ensemble of 200 decision trees with maximum depth 10, minimum samples per leaf 5.

4. **XGBoost**: gradient boosted trees as described above.

4.4 SHAP Feature Attribution

SHAP (SHapley Additive exPlanations) [?] values are computed using `shap.TreeExplainer` on a random sample of 5,000 player-game observations from the 2024 season hold-out, using the final model trained on all prior seasons (2016–2023). Mean absolute SHAP values rank features by their average contribution to prediction magnitude.

4.5 Betting Simulation

A flat-unit betting simulation is conducted over all walk-forward test predictions (2021–2024, $N = 94{,}462$ player-game observations per statistic).

Synthetic market lines. In the absence of historical sportsbook closing lines, we construct three synthetic line proxies of increasing sophistication: (i) the player's 10-game simple rolling average (L_{10}); (ii) a 5-game exponential moving average (EMA_5); and (iii) the linear regression model's prediction for that game, using identical features to XGBoost. The third proxy is the most conservative benchmark: both models see the same information, so any positive XGBoost edge reflects non-linear model capacity rather than information advantage.

Bet signal. A bet is placed when the absolute difference between the XGBoost projection and the synthetic line exceeds a threshold τ:

$$\text{Bet OVER if } \hat{y}_{\text{XGB}} > \ell + \tau, \quad \text{Bet UNDER if } \hat{y}_{\text{XGB}} < \ell - \tau \tag{2}$$

where ℓ is the synthetic line value and $\tau \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5\}$ (in stat units).

Payout structure. Standard -110 juice is assumed: a winning bet returns +0.909 units per unit risked; a losing bet costs -1.0 unit. The break-even win rate is 52.38%. ROI is defined as total profit divided by number of bets placed.

5 Results

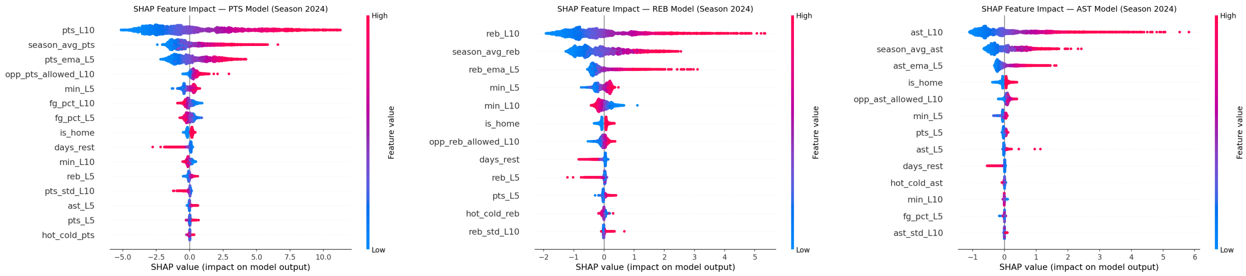
5.1 Prediction Accuracy

Table 3 reports mean absolute error (MAE) and root mean squared error (RMSE) for each model, statistic, and walk-forward fold.

Table 3: Walk-forward MAE by model and statistic (mean across folds 2021–2024). Bold indicates best performance per stat.

Stat	Naive L5	Linear	Rand. Forest	XGBoost	Δ vs Naive
Points	5.082	4.884	4.839	**4.840**	-4.8%
Rebounds	2.070	1.986	1.968	**1.966**	-5.0%
Assists	1.478	1.428	1.413	**1.412**	-4.5%

5.2 SHAP Feature Importance



(a) Points (b) Rebounds (c) Assists

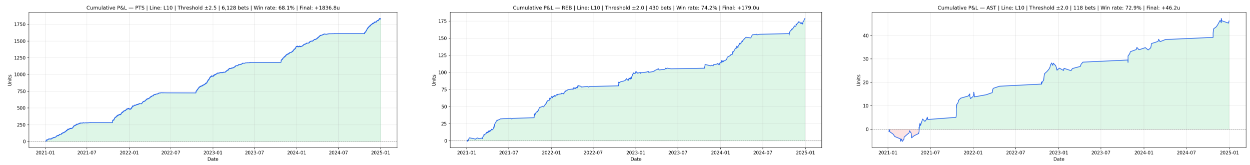
Figure 1: SHAP beeswarm plots for the final XGBoost models evaluated on 5,000 randomly sampled 2024-season observations. Each dot represents one player-game; colour encodes feature value (red = high, blue = low); horizontal position encodes SHAP contribution to the prediction.

5.3 Betting Simulation

Table 4 reports simulated ROI at representative edge thresholds across all three synthetic line definitions.

Table 4: Simulated ROI by statistic, line type, and edge threshold (N = number of bets; break-even ROI = -0.048 at -110 juice). Only thresholds with $N \geq 1{,}000$ reported.

Stat	**Line**	**Threshold**	**N bets**	**ROI**
Points	L_{10}	1.0	42,918	+0.143
	EMA_5	1.0	55,410	+0.198
	LinReg	1.0	17,685	+0.131
Rebounds	L_{10}	1.0	6,416	+0.250
	EMA_5	1.0	17,559	+0.342
	LinReg	1.0	1,525	+0.208
Assists	L_{10}	1.0	2,552	+0.295
	EMA_5	1.0	7,985	+0.339
	LinReg	1.0	793	+0.156



(a) Points (L10 line) (b) Rebounds (L10 line) (c) Assists (L10 line)

Figure 2: Cumulative flat-unit P&L over time (2021–2024) against the L_{10} synthetic line at a 1-unit edge threshold. Each point represents a single bet outcome; shading indicates profit (green) and loss (red) regions.

6 Limitations and Future Work

Synthetic market lines. All betting simulation results are relative to synthetically constructed lines rather than historical sportsbook closing lines. Real prop lines incorporate injury reports, lineup information, travel schedules, and sharp money movement — factors absent from our model. Evaluation against historical closing lines (e.g., via commercial odds data providers) would yield more practically meaningful ROI estimates.

Injury and lineup data. The model does not incorporate real-time roster information. When a star player is absent, the minutes and statistical distribution of teammates shifts substantially. Integrating lineup-adjusted features is a known improvement opportunity in NBA prediction literature [**?**].

Probability calibration. The probability of exceeding the line is estimated via the normal CDF with the player's 10-game standard deviation as the dispersion parameter. This distributional assumption is not validated against empirical frequencies. Calibration plots and post-hoc methods such as Platt scaling [**?**] are left as future work.

Historical era coverage. The training data spans 1951–2024, though league play style has changed substantially. Models trained on all historical data may be distorted by structural shifts in pace, three-point volume, and role specialization. Restricting training to a shorter modern-era window is worth exploring.

7 Conclusion

We presented a gradient boosting regression pipeline for NBA player prop prediction that achieves consistent temporal generalization across four held-out seasons. Walk-forward cross-validation yields mean absolute errors of 4.84 points, 1.97 rebounds, and 1.41 assists — improving over a naive recency baseline by 4.8–5.0%. SHAP analysis confirms that the 10-game rolling average, season expanding mean, and opponent defensive context are the primary predictors, with the exponential moving average contributing non-redundant momentum signal. A betting simulation against a linear regression synthetic line — the most conservative benchmark, as both models share identical features — produces positive ROI across all statistics (PTS: +13.1%, REB: +20.8%, AST: +15.6% at a 1-unit edge threshold), demonstrating that XGBoost's non-linear capacity captures structure a linear model does not.

All code, trained models, and evaluation scripts are available at: `https://github.com/andomo3/nbaPropsPrediction`

References

A Per-Fold MAE Results

Table 5: MAE by model, statistic, and fold year.

Stat	Fold	Naive L5	Linear	RF	XGBoost
Points	2021	4.999	4.805	4.767	4.770
	2022	5.092	4.904	4.856	4.853
	2023	5.124	4.928	4.873	4.877
	2024	5.113	4.898	4.859	4.859
Rebounds	2021	2.069	1.981	1.963	1.963
	2022	2.085	2.001	1.985	1.982
	2023	2.063	1.982	1.963	1.964
	2024	2.061	1.979	1.961	1.957
Assists	2021	1.431	1.381	1.368	1.366
	2022	1.467	1.419	1.403	1.402
	2023	1.488	1.439	1.424	1.423
	2024	1.528	1.475	1.458	1.458

B SHAP Feature Importance Tables

C Full Betting Simulation Results